

Disturbance Rejection Event-Triggered Robust Model Predictive Control for Tracking of Constrained Uncertain Robotic Manipulators

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Abstract—A novel hierarchical control framework combining computed-torque-like control (CTLC) with disturbance-observer-based event-triggered robust model predictive control (DO-ET-RMPC) is proposed for the trajectory tracking control of robotic manipulators with bounded disturbances and state and control input constraints. The CTLC approach is first used to cancel the exact nonlinear dynamics of the original tracking error system to obtain a set of decoupling linear tracking error subsystems, thus reducing the optimization complexity of model predictive control (MPC). The composite DO-ET-RMPC scheme is then developed based on the so-called dual-mode MPC approach to robustly stabilize the tracking error subsystems, which could improve the robustness of MPC and save its computational resources simultaneously. The continuous-time theoretical properties of the DO-ET-RMPC scheme, considering disturbances and state and control input constraints simultaneously, are provided for the first time, including the avoidance of Zeno behavior, robust constraint satisfaction, recursive feasibility, and stability. In the end, the superiorities of the proposed control scheme are verified by the comparative simulations.

Index Terms—Disturbance observer, event-triggered, model predictive control (MPC), robotic manipulators, trajectory tracking.

I. INTRODUCTION

INDUSTRIAL robotic manipulators have been gaining popularity in various automated technology-based manufacturing industries across the globe and can accomplish particular tasks with greater flexibility, accuracy, and efficiency. High-performance motion control is essential to the autonomous operation of industrial robotic manipulators. Recently, a surge of research interest in the motion control of robotic manipulators has trended, focusing primarily on achieving disturbance/uncertainty attenuation, constraint satisfaction, and controller performance optimization [1]. Note that the term “uncertainty” herein denotes the modeling uncertainty, which could be regarded as a class of disturbances, as stated in [2], [3], and [4].

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Not surprisingly, the control accuracy of robotic manipulators would be deteriorated by the effects of various disturbances. To tackle this problem, classic H_∞ control [5], [6], sliding-mode control [7], [8], adaptive control [9], [10], etc. have been proposed to achieve the anti-disturbance control of robotic manipulators. In recent years, the disturbance-observer-based active disturbance rejection technique has been integrated into these classic control approaches, forming the so-called disturbance-observer-based control (DOBC), to further improve the anti-disturbance ability and tracking accuracy of robotic manipulator systems [11], [12], [13], [14], [15]. However, the above DOBC approaches might not be able to directly handle the physical constraints of the manipulators, thus the deterioration of tracking performance might occur when the constraints are violated. To deal with the constraints, the emerging disturbance-observer-based robust model predictive control (DO-RMPC) has also been employed in the motion control of robotic manipulators recently [16], [17], [18], [19]. However, these studies only consider the control input constraints of robotic manipulators.

On the other hand, the high online computational burden of model predictive control (MPC) limited its application in large-scale and fast-varying dynamic systems, including most robotic manipulator systems. In the past decades, event-triggered RMPC (ET-RMPC) has obtained considerable attention due to its ability to reduce the frequency of online solving optimization problems [20], [21], [22], [23], [24], [25], [26]. Specifically, the event-triggering mechanism focuses on detailed system dynamic behaviors and activates control actions only if a specified threshold is violated by a prescribed performance index. For uncertain systems, the prescribed performance index is generally defined as a function concerning the prediction deviation resulting from disturbances. Therefore, in contrast with the regular time-driven RMPC, the ET-RMPC would be able to reduce the number of optimizations thus saving computational resources. However, the research effort on ET-RMPC applied to the motion control of robotic manipulators is also limited. In the recent study [27], an ET-RMPC scheme with learning terminal cost is applied to the tracking control of the uncertain robotic manipulator which is only subject to control input constraints.

Intuitively, it might be constructive to simultaneously integrate event-triggering mechanisms and disturbance observer techniques into RMPC, forming disturbance-observer-based ET-RMPC (DO-ET-RMPC), to achieve the high-performance

tracking control of robotic manipulators. In comparison with the regular DO-RMPC and ET-RMPC schemes, there would be some potential advantages in the DO-ET-RMPC scheme: 1) by introducing the event-triggering mechanism, the online computational burden of the DO-ET-RMPC scheme would be smaller than the regular DO-RMPC scheme; 2) by continuous active disturbance compensation, the robustness of the DO-ET-RMPC scheme would be greater than the regular ET-RMPC scheme; and 3) the prescribed performance index (e.g., the deviation between the actual state and the nominal predicted one in [20] and [21]) in the event-triggering mechanism would be reduced gradually due to active disturbance compensation. The triggering condition would get harder and harder to be met as time goes on, thus the total triggering times would be further reduced. Therefore, it can be predicted that the DO-ET-RMPC scheme would achieve an ideal tradeoff between control performance and computational resource saving. However, to our best knowledge, there might be very limited studies on the DO-ET-RMPC scheme and its application to robotic manipulators. Recently, several similar studies have been reported in [28], [29], and [30]. In [28], the problem of predictive event-driven output feedback disturbance rejection control for unconstrained linear discrete-time systems with disturbances is investigated. Both [29] and [30] design their own DO-ET-RMPC schemes for the trajectory tracking control problems of unmanned surface vehicles with disturbances and control input constraints and the corresponding discrete-time theoretical analyses are also provided. In summary, these control schemes would not be appropriate for the uncertain dynamical systems subject to both state and control input constraints, including most robotic manipulator systems. The continuous-time theoretical analysis of the DO-ET-RMPC scheme is also lacking in the existing results.

Furthermore, it is worth mentioning that, with the gradual enhancement of computer-processing power, the so-called computed-torque-like control (CTLC) is widely used to reduce the nonlinear control problem of robotic manipulators to a set of decoupling linear control problems [2], [3], [31], [32], [33] to reduce the complexity of controller design. By further introducing the parallel computation technique, the real-time performance of the controller would be improved significantly.

According to the above discussions, it would be a challenging problem that integrates event-triggering mechanisms and disturbance observer techniques into MPC to simultaneously improve its robustness and computational efficiency thus improving the tracking performance of robotic manipulators. The difficulties lie in designing an efficient DO-ET-RMPC composite controller, constructing the tightened state and control input constraint sets based on the event-triggering mechanism and compensated disturbances to ensure robust constraint satisfaction, and deriving the rigorous theoretical properties to guarantee the recursive feasibility and stability. To tackle these problems, in this article, a novel hierarchical control framework for the trajectory tracking control problem of constrained uncertain robotic manipulators is investigated. The primary contributions are summarized as follows.

TABLE I
ACRONYMS

DOBC	Disturbance-observer-based control
RMPC	Robust model predictive control
ET-RMPC	Event-triggered robust model predictive control
DO-RMPC	Disturbance-observer-based robust model predictive control
DO-ET-RMPC	Disturbance-observer-based event-triggered robust model predictive control
CTLC	Computed-torque-like control
DOF	Degree of freedom
FISM-RMPC	Filtered integral sliding mode based robust model predictive control
RMSE	Root mean square tracking error
RMSC	Root mean square control input

- 1) A novel hierarchical control scheme combining CTLC with DO-ET-RMPC is designed for the trajectory tracking control of uncertain robotic manipulators with both state and input constraints. Compared with the standard DO-RMPC [16], [17], [18], [19] and ET-RMPC [27] schemes, the DO-ET-RMPC composite controller would improve the robustness of MPC and reduce its triggering times simultaneously by organically integrating a disturbance rejection technique and an event-triggering mechanism into MPC, thus achieving a satisfactory tradeoff between control performance and computational resources.
- 2) Improving upon [29] and [30], the rigorous continuous-time theoretical properties, considering disturbances and state and control input constraints, are thoroughly provided for the first time, including the avoidance of Zeno behavior, robust constraint satisfaction, recursive feasibility, and stability. It is also theoretically demonstrated that, because of the continuous disturbance compensation, the triggering times of the DO-ET-RMPC scheme would be smaller than the ET-RMPC scheme without increasing the conservativeness of the tightened state constraint. Furthermore, comparative simulations are conducted to show the superiorities of the proposed control scheme.

The remainder of this article is organized as follows. In Section II, the manipulator dynamics and the control objective are formulated. Section III presents the scheme of CTLC combined with DO-ET-RMPC. The main theoretical properties are discussed in Section IV. The comparative simulations are provided in Section V. Section VI concludes this article. Notations throughout this article are as follows. The n -dimensional real Euclidean space is denoted by $\mathbf{R}^n$. The set of all non-negative integers is denoted by $\mathbf{I}$. $\mathbf{I}(\mathbf{R})_{[p,q]}$ represent $\{x \in \mathbf{I}(\mathbf{R}) | p \leq x \leq q\}$. $\lambda(\mathbf{Z})$ ($\bar{\lambda}(\mathbf{Z})$) is the minimum (maximum) eigenvalue of the matrix $\mathbf{Z}$. $\text{diag}\{w_i\}$ is a diagonal matrix with diagonal elements $w_i \in \mathbf{R}$, $i \in \mathbf{I}_{[1,\infty)}$. Given a column vector $\mathbf{z}$, $\|\mathbf{z}\|$ and $\|\mathbf{z}\|_{\mathbf{P}} \triangleq \sqrt{\mathbf{z}^T \mathbf{P} \mathbf{z}}$ are denoted as the Euclidean norm and $\mathbf{P}$ -weighted norm, respectively. z_j is the j th element of $\mathbf{z}$. Given a real-valued function $q(t)$, $\sup_{t \in T} |q(t)|$ is the minimum upper bound of $|q(t)|$ for $t \in T$. The main acronyms throughout this article are summarized in Table I.

II. PROBLEM FORMULATION

A. Dynamics of Constrained Uncertain Manipulator

The dynamics of an n -DOF fully actuated serial rigid robotic manipulator can be expressed as

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \mathbf{G}(\boldsymbol{\theta}) + \mathbf{F}(\dot{\boldsymbol{\theta}}) = \boldsymbol{\tau} + \mathbf{d} \quad (1)$$

where $\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}} \in \mathbf{R}^n$ are the joint angular displacements, velocities, and accelerations, respectively. $\mathbf{M}(\boldsymbol{\theta}) \in \mathbf{R}^{n \times n}$ is the symmetric and positive-definite joint-space inertia matrix. $\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) \in \mathbf{R}^{n \times n}$ stands for the centripetal and Coriolis matrix. $\mathbf{G}(\boldsymbol{\theta}) \in \mathbf{R}^n$ denotes the gravitational torques. $\mathbf{F}(\dot{\boldsymbol{\theta}}) \in \mathbf{R}^n$ represents the friction torques. $\boldsymbol{\tau} \in \mathbf{R}^n$ is the control input torques. $\mathbf{d} \in \mathbf{R}^n$ is the bounded lumped disturbances consisting of external disturbances and modeling uncertainties.

The manipulator dynamics (1) can be simplified and expressed in the state space as

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{x}_2 \\ \dot{\mathbf{x}}_2 = \mathbf{M}^{-1}(\mathbf{x}_1)(\boldsymbol{\tau} - \mathbf{H}(\mathbf{x}_1, \mathbf{x}_2)) + \mathbf{d}^* \end{cases} \quad (2)$$

where $\mathbf{x} \triangleq [\mathbf{x}_1^T, \mathbf{x}_2^T]^T \triangleq [\boldsymbol{\theta}^T, \dot{\boldsymbol{\theta}}^T]^T \in \mathbf{R}^{2n}$. $\mathbf{H}(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{C}(\mathbf{x}_1, \mathbf{x}_2)\mathbf{x}_2 + \mathbf{G}(\mathbf{x}_1) + \mathbf{F}(\mathbf{x}_2) \in \mathbf{R}^n$. $\mathbf{d}^* \triangleq \mathbf{M}^{-1}(\mathbf{x}_1)\mathbf{d} \in \mathbf{R}^n$ is the equivalent lumped disturbances. Referring to [1], the joint and disturbance constraint sets are given by

$$\mathbf{x} \in \mathcal{S} \triangleq \left\{ \mathbf{x} \in \mathbf{R}^{2n} \mid |x_{2j}| \leq s_j, j \in \mathbf{I}_{[1,n]} \right\} \quad (3a)$$

$$\boldsymbol{\tau} \in \mathcal{C} \triangleq \left\{ \boldsymbol{\tau} \in \mathbf{R}^n \mid |\tau_j| \leq c_j, j \in \mathbf{I}_{[1,n]} \right\} \quad (3b)$$

$$\mathbf{d}^* \in \mathcal{D} \triangleq \left\{ \mathbf{d}^* \in \mathbf{R}^n \mid |d_j^*| \leq \xi_{1j}, |\dot{d}_j^*| \leq \xi_{2j}, j \in \mathbf{I}_{[1,n]} \right\} \quad (3c)$$

where $\mathcal{S} \subset \mathbf{R}^{2n}$, $\mathcal{C} \subset \mathbf{R}^n$, and $\mathcal{D} \subset \mathbf{R}^n$ are the convex sets including the origin as an interior point. x_{ij} , τ_j , and d_j^* are the j th elements of $\mathbf{x}_i$, $\boldsymbol{\tau}$, and $\mathbf{d}^*$, respectively. $s_j, c_j, \xi_{1j}, \xi_{2j} > 0$ are the known constants.

For the serial manipulators with only revolute joints, the following dynamic characteristic holds [31].

Property 1: The centrifugal and Coriolis torques, gravitational torques, and friction torques are norm bounded for all possible joint angular displacements and velocities, which implies that there exist the constants $h_j > 0$ such that

$$\|(\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\dot{\boldsymbol{\theta}} + \mathbf{G}(\boldsymbol{\theta}) + \mathbf{F}(\dot{\boldsymbol{\theta}}))_j\| \leq h_j, j \in \mathbf{I}_{[1:n]}. \quad (4)$$

B. Control Objective

In this study, the control objective is to drive a manipulator, in the presence of bounded disturbances, to track a desired time-varying trajectory in the joint space, while satisfying (3a) and (3b).

The dynamics (disturbance-free) of the desired trajectory is generated by a second-order integrator as

$$\begin{cases} \dot{\mathbf{x}}_{d1} = \mathbf{x}_{d2} \\ \dot{\mathbf{x}}_{d2} = \mathbf{u}_d \end{cases} \quad (5)$$

where $\mathbf{x}_d \triangleq [\mathbf{x}_{d1}^T, \mathbf{x}_{d2}^T]^T \in \mathcal{S}$. $\mathbf{x}_{d1}$ is assumed to be twice continuously differentiable on $\mathbf{R}_{[t_0, t_p]}$, where t_0 and t_p are the initial and terminal time instants, respectively. $\mathbf{u}_d \triangleq \mathbf{M}^{-1}(\mathbf{x}_{d1})(\boldsymbol{\tau}_d - \mathbf{H}(\mathbf{x}_{d1}, \mathbf{x}_{d2}))$, where $\boldsymbol{\tau}_d \in \mathcal{C}$ is the reference

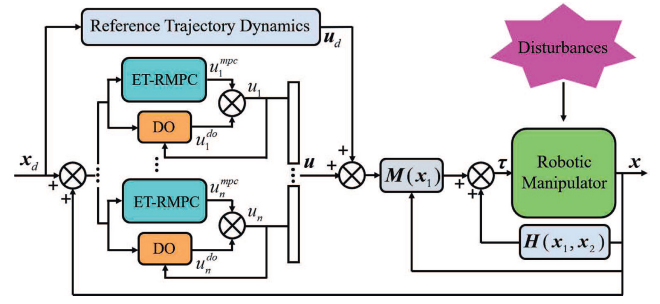


Fig. 1. CTLC combined with DO-ET-RMPC.

control input. The j th elements of $\mathbf{x}_{di}$ and $\mathbf{u}_d$ are denoted as x_{dij} and u_{dj} , respectively.

Then, by subtracting (5) from (2), the tracking error dynamics in the state space can be obtained as

$$\begin{cases} \dot{\tilde{\mathbf{e}}}_1 = \tilde{\mathbf{e}}_2 \\ \dot{\tilde{\mathbf{e}}}_2 = \mathbf{M}^{-1}(\mathbf{x}_1)(\boldsymbol{\tau} - \mathbf{H}(\mathbf{x}_1, \mathbf{x}_2)) + \mathbf{d}^* - \mathbf{u}_d \end{cases} \quad (6)$$

where $\tilde{\mathbf{e}} \triangleq [\tilde{\mathbf{e}}_1^T, \tilde{\mathbf{e}}_2^T]^T \triangleq \mathbf{x} - \mathbf{x}_d \in \mathbf{R}^{2n}$ is the tracking error state. e_{ij} and d_j^* are the j th elements of $\tilde{\mathbf{e}}_i$ and $\mathbf{d}^*$, respectively.

III. CTLC COMBINED WITH DO-ET-RMPC

The control scheme of CTLC combined with DO-ET-RMPC is graphically rendered in Fig. 1.

A. CTLC Approach

Then, the so-called CTLC is used to decouple the dynamics (6) and cancel its exact nonlinear component. The control input is chosen as

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{x}_1)(\mathbf{u} + \mathbf{u}_d) + \mathbf{H}(\mathbf{x}_1, \mathbf{x}_2) \quad (7)$$

where $\mathbf{u} \triangleq [u_1, u_2, \dots, u_n]^T \in \mathbf{R}^n$ is the auxiliary control input. By substituting (7) into (6), the tracking error system for the j th joint in the compact matrix form can be seen as

$$\dot{\mathbf{e}}_j = \mathbf{A}\mathbf{e}_j + \mathbf{B}(u_j + d_j^*) \quad (8)$$

where $\mathbf{e}_j = [e_{1j}, e_{2j}]^T \in \mathcal{S}_{ej} \subset \mathbf{R}^2$ and $u_j \in \mathcal{C}_{ej} \subset \mathbf{R}$ are the tracking error state and control input, respectively, of the joint j . $\mathcal{S}_{ej}$ and $\mathcal{C}_{ej}$ are the convex sets containing the origin in their interior. The specific expressions of $\mathcal{S}_{ej}$ and $\mathcal{C}_{ej}$ are obtained by following the Minkowski set subtraction [1], [33]. Specifically, from (3a) and $x_{2j} = x_{d2j} + e_{2j}$, it is seen that if $|e_{2j}| \leq s_j - |x_{d2j}|$ holds, the strict satisfaction of the original constraint (3a) is guaranteed, where $|x_{d2j}| \triangleq \sup_{t \in \mathbf{R}_{[t_0, t_p]}} |x_{d2j}(t)|$. Furthermore, from (3b), (4), and (7), we have that if $|\mathbf{M}_j(\mathbf{x}_1)(\mathbf{u} + \mathbf{u}_d)| \leq c_j - h_j$ holds, the strict satisfaction of the original constraint (3b) is guaranteed, where $\mathbf{M}_j(\mathbf{x}_1)$ is the j th row vector of $\mathbf{M}(\mathbf{x}_1)$. In this article, the inertia matrix $\mathbf{M}(\mathbf{x}_1)$ is chosen as a constant diagonal matrix, denoted as $\mathbf{M}_c$, which is widely used in the design of the CTLC-based robust controllers for manipulators [2], [3], [32]. Then, the following n inequalities can be obtained that:

$$h_j - c_j \leq m_{jj}(u_j + u_{dj}) \leq c_j - h_j, j \in \mathbf{I}_{[1,n]} \quad (9)$$

where m_{jj} is the j th diagonal element of $\mathbf{M}_c$.

Based on the above, the constraint sets $\mathcal{S}_{ej}$ and $\mathcal{C}_{ej}$ are chosen that

$$\mathcal{S}_{ej} \triangleq \left\{ \mathbf{e}_j \in \mathbf{R}^2 \mid |e_{2j}| \leq s_j - \kappa_{1j} \overline{|x_{d2j}|}, j \in \mathbf{I}_{[1,n]} \right\} \quad (10a)$$

$$\mathcal{C}_{ej} \triangleq \left\{ \mathbf{u}_j \in \mathbf{R} \mid |u_j| \leq v_j - \kappa_{2j} \overline{|u_{dj}|}, j \in \mathbf{I}_{[1,n]} \right\} \quad (10b)$$

where $\overline{|u_{dj}|} \triangleq \sup_{t \in \mathbf{R}_{[t_0, t_p]}} |u_{dj}(t)|$ and $v_j \triangleq (c_j - h_j)/m_{jj}$. $\kappa_{1j}, \kappa_{2j} \geq 1$ are the preassigned overshoot factors.

Remark 1: Similar to the idea of the prescribed performance control [13], the overshoot factors κ_{1j}, κ_{2j} could be increased to suppress the overshoot in tracking thus preserving the transient and steady-state performance of the close-loop system. However, larger κ_{1j} and κ_{2j} would cause a longer prediction horizon thus increasing the computational burden of MPC. The tradeoff between control performance and the computational burden needs to be considered when designing κ_{1j} and κ_{2j} . In this article, both κ_{1j} and κ_{2j} are set to 1 to ensure weakly conservative constraint sets.

Remark 2: By virtue of (7), the stability analysis of the nonlinear coupling tracking error system (6) is reduced to the analyses of n decoupled linear tracking error subsystems (8). For the sake of simplicity, the subsequent descriptions only focus on the j th subsystem.

B. DO-ET-RMPC Scheme

The DO-ET-RMPC scheme, with active disturbance rejection and computational resource saving, is presented in this section. According to the basic idea of DOBC [4], the control input u_j herein is designed as

$$u_j \triangleq u_j^{do} + u_j^{mpc} \quad (11)$$

where u_j^{do} and u_j^{mpc} are the disturbance compensation input and ET-RMPC input, respectively.

1) *Design of u_j^{do} :* The disturbance observer is designed to estimate the disturbances d_j^* in (8), as

$$\begin{cases} \dot{v}_j = L_j \hat{d}_j^* + L_j \mathbf{B}^\dagger (\mathbf{A} \mathbf{e}_j + \mathbf{B} u_j) \\ \hat{d}_j^* = v_j - L_j \mathbf{B}^\dagger \mathbf{e}_j \end{cases} \quad (12)$$

where $\mathbf{B}^\dagger \triangleq (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T$ is the left pseudoinverse of $\mathbf{B}$. $L_j < 0$ is the observer gain. v_j is the auxiliary variable. $\hat{d}_j^*$ is the disturbance estimate with $\hat{d}_j^*(t_0) = 0$.

By defining the disturbance estimation error as $\tilde{d}_j^* \triangleq d_j^* - \hat{d}_j^*$, it can be obtained that

$$\dot{\tilde{d}}_j^* = \dot{d}_j^* + L_j \tilde{d}_j^*. \quad (13)$$

Considering d_j^* being imposed in the input channel, the disturbance compensation input can be designed as

$$u_j^{do} \triangleq -\hat{d}_j^*. \quad (14)$$

Then, by substituting (11) and (14) into (8), the resulting closed-loop system with the residual disturbances is obtained as

$$\dot{\mathbf{e}}_j = \mathbf{A} \mathbf{e}_j + \mathbf{B} (u_j^{mpc} + \tilde{d}_j^*). \quad (15)$$

The following lemma is useful for deriving the robust tightened constraint sets and event-triggering condition.

Lemma 1: In terms of the disturbance estimation dynamics (12) and the disturbance estimation error dynamics (13), it can be obtained that

$$\|\tilde{d}_j^*\| \leq \left(\xi_{1j} + \frac{\xi_{2j}}{L_j} \right) e^{L_j(t-t_0)} - \frac{\xi_{2j}}{L_j} \quad (16)$$

$$\|\dot{\tilde{d}}_j^*\| \leq \xi_{1j} \left(1 - e^{L_j(t-t_0)} \right). \quad (17)$$

Proof: By solving (12) and (13), it is seen that

$$\tilde{d}_j^*(t) = e^{L_j(t-t_0)} \tilde{d}_j^*(t_0) + \int_{t_0}^t e^{L_j(t-\tau)} \dot{\tilde{d}}_j^*(\tau) d\tau \quad (18)$$

$$\dot{\tilde{d}}_j^*(t) = L_j \hat{d}_j^*(t) - L_j d_j^*(t). \quad (19)$$

According to (18), it can be obtained that

$$\begin{aligned} \|\tilde{d}_j^*(t)\| &\leq \|e^{L_j(t-t_0)} \tilde{d}_j^*(t_0)\| + \left\| \int_{t_0}^t e^{L_j(t-s)} \dot{\tilde{d}}_j^*(s) ds \right\| \\ &\leq \xi_{1j} e^{L_j(t-t_0)} + \xi_{2j} \int_{t_0}^t e^{L_j(t-s)} ds \\ &= \left(\xi_{1j} + \frac{\xi_{2j}}{L_j} \right) e^{L_j(t-t_0)} - \frac{\xi_{2j}}{L_j}. \end{aligned} \quad (20)$$

Further, according to (19), it is seen that

$$\begin{aligned} \|\dot{\tilde{d}}_j^*(t)\| &= \left\| - \int_{t_0}^t e^{L_j(t-s)} L_j \tilde{d}_j^*(s) ds \right\| \\ &\leq \xi_{1j} e^{L_j t} \int_{t_0}^t \| -L_j e^{-L_j s} \| ds \\ &= \xi_{1j} \left(1 - e^{L_j(t-t_0)} \right). \end{aligned} \quad (21)$$

Remark 3: It can be seen from (16) that, by choosing an appropriate observer gain L_j such that $\xi_{1j} + (\xi_{2j}/L_j) \geq 0$ and $L_j \gg \xi_{2j}$, $\|\tilde{d}_j^*\|$ can be decreasing monotonically, and $\lim_{t \rightarrow \infty} \|\tilde{d}_j^*\|$ can be made arbitrary small. In addition, from (17), we can also obtain that $\|\dot{\tilde{d}}_j^*(t)\| < \xi_{1j}, \forall t > t_0$.

2) *Design of u_j^{mpc} :* The residual disturbances $\tilde{d}_j^*$ in (15) are rejected by the ET-RMPC input.

a) *Optimization problem:* Start from the nominal system (disturbance-free) of the perturbed system (15), as

$$\dot{\bar{\mathbf{e}}}_j(t) = \mathbf{A} \bar{\mathbf{e}}_j(t) + \mathbf{B} u_j^{mpc}(t) \quad (22)$$

where $\bar{\mathbf{e}}_j(t)$ is the nominal tracking error state. Denoting the event-triggered time instant sequence $\{t_k\}, k \in \mathbf{I}_{[0, \infty)}$, a finite-horizon optimization problem at the time instant t_k , with the prediction horizon $T \geq 0$, is formulated as follows.

Problem 1:

$$\bar{u}_j^0(\cdot|t_k) = \arg \min_{\bar{u}_j(\cdot|t_k) \in \bar{\mathcal{C}}_{ej}} \mathcal{J}_j(\bar{u}_j(\cdot|t_k), \mathbf{e}_j(t_k)) \quad (23)$$

subject to

$$\bar{\mathbf{e}}_j(t_k|t_k) = \mathbf{e}_j(t_k) \quad (24a)$$

$$\dot{\bar{\mathbf{e}}}_j(t|t_k) = \mathbf{A} \bar{\mathbf{e}}_j(t|t_k) + \mathbf{B} \bar{u}_j(t|t_k), t \in \mathbf{R}_{[t_k, t_k+T]} \quad (24b)$$

$$\bar{\mathbf{e}}_j(t|t_k) \in \bar{\mathcal{S}}_{ej}(t - t_k) \quad (24c)$$

$$\bar{\mathbf{e}}_j(t_k + T|t_k) \in \Omega_{2j}(\epsilon_j). \quad (24d)$$

The cost function is defined as

$$\mathcal{J}_j(\bar{u}_j(\cdot|t_k), \mathbf{e}_j(t_k)) \triangleq \int_{t_k}^{t_k+T} \mathcal{L}_j(\bar{\mathbf{e}}_j(t|t_k), \bar{\mathbf{u}}_j(t|t_k)) dt + \mathcal{V}_j(\bar{\mathbf{e}}_j(t_k + T|t_k)) \quad (25)$$

where $\mathcal{L}_j(\cdot, \cdot) \triangleq \|\cdot\|_{\mathbf{Q}_j}^2 + \|\cdot\|_{\mathbf{R}_j}^2$ is the stage cost function. $\mathcal{V}_j(\cdot) \triangleq \|\cdot\|_{\mathbf{P}_j}^2$ is the terminal cost function. $R_j > 0$ and that $\mathbf{Q}_j, \mathbf{P}_j \in \mathbf{R}^{2 \times 2}$ are symmetric positive-definite matrixes. $\bar{\mathbf{u}}_j(\cdot|t_k)$ and $\bar{\mathbf{e}}_j(\cdot|t_k)$ are the feasible control input and its corresponding predicted state trajectories, respectively. The optimal control input and its corresponding optimal predicted state trajectories to Problem 1 are denoted as $\bar{\mathbf{u}}_j^0(\cdot|t_k)$ and $\bar{\mathbf{e}}_j^0(\cdot|t_k)$, respectively. The tightened state constraint set $\bar{\mathcal{S}}_{ej}$, control constraint set $\bar{\mathcal{C}}_{ej}$, and terminal state constraint set $\Omega_j(\epsilon_j)$ are defined, respectively, as

$$\bar{\mathcal{S}}_{ej}(t - t_k) \triangleq \left\{ \bar{\mathbf{e}}_j \in \mathbf{R}^2 \mid \|\bar{\mathbf{e}}_{2j}\| \leq s_j - \kappa_{1j} \overline{|x_{d2j}|} - \bar{D}_j^* e^{t-t_k} + \bar{D}_j^* \right. \\ \left. j \in \mathbf{I}_{[1,n]} \right\} \quad (26a)$$

$$\bar{\mathcal{C}}_{ej} \triangleq \left\{ \bar{\mathbf{u}}_j \in \mathbf{R} \mid \|\bar{\mathbf{u}}_j\| \leq v_j - \kappa_{2j} \overline{|u_{dj}|} - \xi_{1j}, j \in \mathbf{I}_{[1,n]} \right\} \quad (26b)$$

$$\Omega_j(\epsilon_j) \triangleq \left\{ \bar{\mathbf{e}}_j \in \mathbf{R}^2 \mid \|\bar{\mathbf{e}}_j\|_{\mathbf{P}_j}^2 \leq \epsilon_j^2, \epsilon_j > 0, j \in \mathbf{I}_{[1,n]} \right\}. \quad (26c)$$

Note that the tightened constraint sets $\bar{\mathcal{S}}_{ej}$ and $\bar{\mathcal{C}}_{ej}$ with the associated parameter $\bar{D}_j^*$ are derived according to Theorem 1. The parameters $\mathbf{Q}_j$, R_j , and $\mathbf{P}_j$ herein are determined by using Lemma 2 while ϵ_j is determined by Theorem 2.

For the stabilizable system (22), there exists a state-feedback gain $\mathbf{K}_j$ such that $\mathbf{A}_{jk} \triangleq \mathbf{A}_j + \mathbf{B}_j \mathbf{K}_j$ is Hurwitz. Then, a well-known result is recalled as follows [34], [35].

Lemma 2: For the system (22), given a symmetric positive-definite matrix $\mathbf{Q}_j$ and a constant R_j , there exist a constant ϵ_j and a symmetric positive-definite matrix $\mathbf{P}_j$ such that: 1) $\Omega_j(\gamma_j) \triangleq \{\bar{\mathbf{e}}_j \in \mathbf{R}^2 \mid \|\bar{\mathbf{e}}_j\| \leq \gamma_j^2, \gamma_j > 0, j \in \mathbf{I}_{[1,n]}\}$ is control invariant set with the control input $\mathbf{K}_j \bar{\mathbf{e}}_j \in \bar{\mathcal{C}}_{ej}$ and 2) $\dot{\mathbf{V}}_j(\bar{\mathbf{e}}_j) \leq -\|\bar{\mathbf{e}}_j\|_{\mathbf{Q}_j}^2$ for $\bar{\mathbf{e}}_j \in \Omega_j(\gamma_j)$, where $\mathbf{Q}_j^* \triangleq \mathbf{Q}_j + \mathbf{K}_j^T R_j \mathbf{K}_j$.

b) Event-triggering mechanism: Note that the nominal optimal predicted state trajectory $\bar{\mathbf{e}}_j^0(t|t_k)$ of the system (22) under the control input $\bar{\mathbf{u}}_j^0(\cdot|t_k)$ would not coincide with the actual state trajectory $\mathbf{e}_j(t)$ of the system (8) under the composite control input (11), for $t \in \mathbf{R}_{[t_k, t_k+T]}$. Motivated by this deviation of the trajectories, the event-triggering condition of the DO-ET-RMPC scheme can be defined as

$$t'_{k+1} \triangleq \inf_{t > t_k} \left\{ t \mid \|\mathbf{e}_j(t|t_k) - \bar{\mathbf{e}}_j^0(t|t_k)\| = \varpi_j \right\} \quad (27a)$$

$$t_{k+1} \triangleq \min\{t'_{k+1}, t_k + T\} \quad (27b)$$

where $\varpi_j = \bar{D}_j^* (e^{\beta_j T} - 1)$ is the triggering threshold with $\beta_j \in \mathbf{R}_{(0,1)}$. $\bar{D}_j^*$ will be defined in the proof of Theorem 1. Assume that Problem 1 is triggered automatically at t_0 . To ensure that the proposed DO-ET-RMPC scheme can avoid the so-called Zeno behavior, the following theorem is presented.

Theorem 1: For the perturbed system (8), if the time instant sequence $\{t_k\}$ is generated by the triggering condition (27), the upper and lower bounds of the interexecution time are guaranteed as $\sup_{k \in \mathbf{I}_{[0,\infty)}} \{t_{k+1} - t_k\} \leq T$ and $\inf_{k \in \mathbf{I}_{[0,\infty)}} \{t_{k+1} - t_k\} \geq \beta_j T$, respectively.

Algorithm 1 DO-ET-RMPC

```

while  $\mathbf{e}_j(t) \notin \Omega_j(\gamma_j)$  do
  if  $k = 0$  then
    Solve Problem 1 at  $t_0$ 
     $\mathbf{u}_j(t) \leftarrow (\mathbf{u}_j^{do}(t) \leftarrow -\hat{\mathbf{d}}_j^*(t)) + (\mathbf{u}_j^{mpc}(t) \leftarrow \bar{\mathbf{u}}_j^0(t|t_0))$ 
  end if
  while  $t_{k+1}$  is not triggered do
     $\mathbf{u}_j(t) \leftarrow (\mathbf{u}_j^{do}(t) \leftarrow -\hat{\mathbf{d}}_j^*(t)) + (\mathbf{u}_j^{mpc}(t) \leftarrow \bar{\mathbf{u}}_j^0(t|t_k))$ 
  end while
   $k \leftarrow k + 1$ 
  Solve Problem 1 at  $t_{k+1}$ 
end while
 $\mathbf{u}_j(t) \leftarrow (\mathbf{u}_j^{do}(t) \leftarrow -\hat{\mathbf{d}}_j^*(t)) + (\mathbf{u}_j^{mpc}(t) \leftarrow \mathbf{K}_j \mathbf{e}_j(t))$ 

```

Proof: First, the upper bound of the interexecution time can be seen directly from (27b). Second, considering the system (8) under the composite control input (11) and the system (22) under the control input $\bar{\mathbf{u}}_j^0(\cdot|t_k)$, for $t \in \mathbf{R}_{[t_k, t_k+T]}$, it is seen that

$$\dot{\mathbf{e}}_j(t) - \dot{\bar{\mathbf{e}}}_j^0(t|t_k) = \mathbf{A}(\mathbf{e}_j(t) - \bar{\mathbf{e}}_j^0(t|t_k)) + \mathbf{B} \tilde{\mathbf{d}}_j^*(t). \quad (28)$$

Then, it can be obtained that

$$\begin{aligned} \|\mathbf{e}_j(t) - \bar{\mathbf{e}}_j^0(t|t_k)\| &= \left\| \int_{t_k}^t e^{\mathbf{A}(t-s)} \mathbf{B} \tilde{\mathbf{d}}_j^*(s) ds \right\| \\ &\leq \int_{t_k}^t e^{t-s} \|\tilde{\mathbf{d}}_j^*(s)\| ds \\ &\leq D_j^*(t_k, t) (e^{t-t_k} - 1) \end{aligned} \quad (29)$$

where $D_j^*(t_k, t) \triangleq \sup_{s \in \mathbf{R}_{[t_k, t]}} \|\tilde{\mathbf{d}}_j^*(s)\|$. Further, it can be obtained that $\|\mathbf{e}_j(t) - \bar{\mathbf{e}}_j^0(t|t_k)\| \leq \bar{D}_j^* (e^{t-t_k} - 1)$ where $\bar{D}_j^* \triangleq D_j^*(t_0, \infty)$ [36]. Then, in terms of (27a), it is seen that $t'_{k+1} \geq t_k + \beta_j T$. By using (27b), it is obtained that $\inf_{k \in \mathbf{I}_{[0,\infty)}} \{t_{k+1} - t_k\} \geq \beta_j T$. The proof is completed. ■

Remark 4: In view of Remark 3, when choosing an appropriate L_j such that $\xi_{1j} + (\xi_{2j}/L_j) \geq 0$, it is seen that the upper bound of $\|\tilde{\mathbf{d}}_j^*\|$ is the same as that of $\|\mathbf{d}_j^*\|$. Thus, the triggering condition and tightened state constraint set designed herein based on the upper bound of $\|\tilde{\mathbf{d}}_j^*\|$ will be the same as those designed based on the upper bound of $\|\mathbf{d}_j^*\|$ in the regular ET-RMPC scheme [21]. Nevertheless, the unique feature of the proposed control scheme is that the part of the control effort is used to compensate disturbances and thus the original disturbances are gradually becoming smaller, which would enable not only higher control performance but also fewer triggering times than the regular ET-RMPC scheme.

Remark 5: In this article, the classic state-deviation-based event-triggering mechanism [20], [21] is introduced into the DO-ET-RMPC scheme. The main objective herein is to verify the effectiveness of this novel control scheme. The discussion of the impact of different event-triggering mechanisms on control performance deserves further study.

3) DO-ET-RMPC Algorithm: With reference to the so-called dual-mode MPC strategy in [35], the DO-ET-RMPC algorithm is described in Algorithm 1.

IV. THEORETICAL ANALYSIS

A. Recursive Feasibility

The sufficient conditions for ensuring the recursive feasibility of the DO-ET-RMPC algorithm are presented.

Theorem 2: Suppose that Problem 1 admits a feasible solution at t_0 , then Algorithm 1 is always feasible for $t_k > t_0$, if the following conditions hold:

$$\sqrt{\bar{\lambda}(\mathbf{P}_j)} \varpi_j e^{-\beta_j T + T} \leq \gamma_j - \epsilon_j \quad (30)$$

$$T \geq 2 \frac{\bar{\lambda}(\mathbf{P}_j)}{\underline{\lambda}(\mathbf{Q}_j^*) \beta_j} \ln \frac{\gamma_j}{\epsilon_j} \quad (31)$$

$$E_j \leq s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^T + \bar{D}_j^* \quad (32)$$

where $E_j = \left(\gamma_j / \sqrt{P_j(22)} - \frac{P_j^2(12)}{P_j(11)} \right)$. $P_j(mn)$ is the element at the m th row and n th column of the matrix $\mathbf{P}_j$.

Proof: A candidate control input to Problem 1 at t_{k+1} is constructed as

$$\tilde{u}_j(t|t_{k+1}) = \begin{cases} \tilde{u}_j^0(t|t_k), & t \in \mathbf{R}_{[t_{k+1}, t_k+T]}, \\ \mathbf{K}_j \tilde{e}_j(t|t_{k+1}), & t \in \mathbf{R}_{[t_k+T, t_{k+1}+T]} \end{cases} \quad (33)$$

where $\tilde{e}_j(t|t_{k+1})$ is the predicted state trajectory generated by $\tilde{u}_j(t|t_{k+1})$, $\tilde{e}_j(t_{k+1}|t_{k+1}) = e_j(t_{k+1})$.

First, for $t \in \mathbf{R}_{[t_{k+1}, t_k+T]}$, by using (29), it is seen that

$$\begin{aligned} \|\tilde{e}_j(t|t_{k+1}) - \tilde{e}_j^0(t|t_k)\| &= \|e^{A(t-t_{k+1})} (e_j(t_{k+1}) - \tilde{e}_j^0(t_{k+1}|t_k))\| \\ &\leq \varpi_j e^{(t-t_{k+1})}. \end{aligned} \quad (34)$$

Substituting $t = t_k + T$ into (34) yields that $\|\tilde{e}_j(t_k + T|t_{k+1}) - \tilde{e}_j^0(t_k + T|t_k)\| \leq \varpi_j e^{(t_k+T-t_{k+1})}$. By using (30), it can be seen that $\|\tilde{e}_j(t_k + T|t_{k+1})\|_{\mathbf{P}_j} \leq \epsilon_j + \sqrt{\bar{\lambda}(\mathbf{P}_j)} \varpi_j e^{(t_k+T-t_{k+1})} \leq \gamma_j$. For $t \in \mathbf{R}_{[t_k+T, t_{k+1}+T]}$, it is seen that $\dot{\mathcal{V}}_j(\tilde{e}_j(t|t_{k+1})) \leq -(\underline{\lambda}(\mathbf{Q}_j^*)/\bar{\lambda}(\mathbf{P}_j)) \mathcal{V}_j(\tilde{e}_j(t|t_{k+1}))$. By virtue of the comparison principle, we have $\mathcal{V}_j(\tilde{e}_j(t|t_{k+1})) \leq \mathcal{V}_j(\tilde{e}_j(t_k + T|t_{k+1})) e^{-(\underline{\lambda}(\mathbf{Q}_j^*)/\bar{\lambda}(\mathbf{P}_j))(t-t_k-T)}$. By using (31), it can be seen that $\|\tilde{e}_j(t_{k+1} + T|t_{k+1})\|_{\mathbf{P}_j} \leq \gamma_j e^{-(\underline{\lambda}(\mathbf{Q}_j^*)/2\bar{\lambda}(\mathbf{P}_j))\beta_j T} \leq \epsilon_j$.

Second, it is seen from (33) that $\tilde{u}_j(t|t_{k+1}) \in \bar{\mathcal{C}}_{ej}$, $t \in \mathbf{R}_{[t_{k+1}, t_k+T]}$. For $t \in \mathbf{R}_{[t_k+T, t_{k+1}+T]}$, the control input $\mathbf{K}_j \tilde{e}_j(t|t_{k+1})$ is adopted to remain the system state in $\Omega_j(\gamma_j)$. Thus, it follows Lemma 2 that $\tilde{u}_j(t|t_{k+1}) \in \bar{\mathcal{C}}_{ej}$, $t \in \mathbf{R}_{[t_k+T, t_{k+1}+T]}$.

Third, according to (34), for $t \in \mathbf{R}_{[t_{k+1}, t_k+T]}$, it can be seen that $\|\tilde{e}_{2j}(t|t_{k+1})\| \leq s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^{t-t_k} + \bar{D}_j^* + \varpi_j e^{(t-t_{k+1})} = s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^{t-t_k} + \bar{D}_j^* + \bar{D}_j^* (e^{\beta_j T} - 1) e^{(t-t_{k+1})}$. By using Theorem 1 again, we have $\|\tilde{e}_{2j}(t|t_{k+1})\| \leq s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^{t-t_{k+1}} + \bar{D}_j^*$. Further, for $t \in \mathbf{R}_{[t_k+T, t_{k+1}+T]}$, it is seen that $\tilde{e}_j(t|t_{k+1}) \in \Omega_j(\gamma_j)$, which implies that $\|\tilde{e}_{2j}(t|t_{k+1})\| \leq E_j$. As shown in Fig. 2, $2E_j$ is the length of the minimum enclosing rectangle of the region $\Omega_j(\gamma_j)$. Then, it follows (32) that $|\tilde{e}_{2j}(t|t_{k+1})| \leq s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^T + \bar{D}_j^* \leq s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^{t-t_{k+1}} + \bar{D}_j^*$.

Based on the above three arguments, the candidate control input $\tilde{u}_j(t|t_{k+1})$ is proved to be a feasible solution to Problem 1, thus the proof is completed. ■

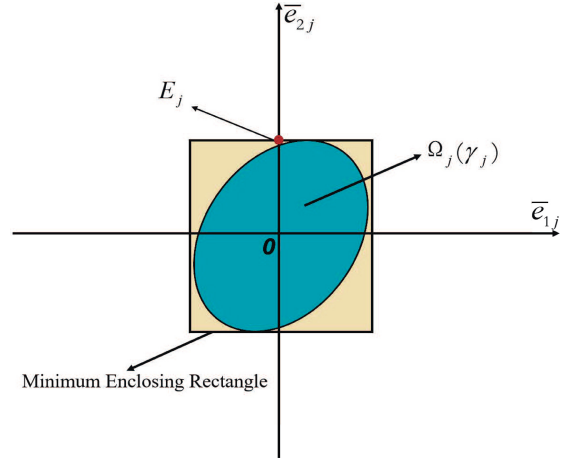


Fig. 2. Minimum enclosing rectangle of green region $\Omega_j(\gamma_j)$.

Remark 6: According to Theorem 2, it can be seen that the recursive feasibility of Algorithm 1 would rely on the minimum interexecution time $\beta_j T$ and the residual disturbance upper bound $\bar{D}_j^*$. In particular, a larger β_j would lead to the greater disturbance tolerability of the triggering mechanism thus reducing the triggering times, however, the control performance might be also sacrificed due to less frequent sampling actions. Moreover, a larger T would achieve better control performance since the longer state evolution is considered in online optimization computation, whereas it would consume more computational resources.

B. Closed-Loop Stability

The closed-loop stability of the j th tracking error subsystem (8) under the proposed DO-ET-RMPC scheme is investigated here. According to Remark 3, the closed-loop stability of the original tracking error system (6) under the control input (7) is also guaranteed.

Theorem 3: For the j th tracking error subsystem under the control input generated by Algorithm 1, assuming the conditions in Theorem 2 are satisfied, if the following condition holds:

$$\begin{aligned} &\bar{\lambda}(\mathbf{Q}_j)(1 - \beta_j)T \varpi_j e^{(T-\beta_j T)} (\varpi_j e^{(T-\beta_j T)} + 2\bar{\Psi}_j) \\ &+ \sqrt{\bar{\lambda}(\mathbf{P}_j)} \varpi_j e^{T-\beta_j T} (\gamma_j + \epsilon_j) \leq \underline{\lambda}(\mathbf{Q}_j) \beta_j T \left(\frac{\gamma_j}{\sqrt{\bar{\lambda}(\mathbf{P}_j)}} - \varpi_j \right)^2 \end{aligned} \quad (35)$$

where $\bar{\Psi}_j \triangleq \sqrt{(2\pi)^2 + (s_j - \kappa_{1j} \bar{x}_{d2j} - \bar{D}_j^* e^{\beta_j T} + \bar{D}_j^*)^2}$, then the tracking error state $e_j(t)$ will enter to the set $\Omega_j(\gamma_j)$ in finite time. Further, when $e_j(t) \in \Omega_j(\gamma_j)$, it will remain there if the following condition holds that:

$$\frac{2\bar{\lambda}(\mathbf{P}_j) \|\sqrt{\bar{\mathbf{P}}_j}\| \bar{D}_j^*}{\underline{\lambda}(\mathbf{Q}_j^*)} < \gamma_j. \quad (36)$$

Proof: This theorem is proved by two steps. In the first step, we show that if $e_j(t_0) \notin \Omega_j(\gamma_j)$, it will enter $\Omega_j(\gamma_j)$ in finite time. In the second step, we prove that if $e_j(t_0) \in \Omega_j(\gamma_j)$, the closed system under the control input $\mathbf{K}_j e_j(t)$ will remain stable in $\Omega_j(\gamma_j)$.

First, we chose a function as

$$\Delta \mathcal{J}_j \triangleq \mathcal{J}_j(\tilde{u}_j^0(t|t_{k+1}), \mathbf{e}_j(t_{k+1})) - \mathcal{J}_j(\tilde{u}_j^0(t|t_k), \mathbf{e}_j(t_k)). \quad (37)$$

According to the suboptimality of (33), it is seen that

$$\begin{aligned} \Delta \mathcal{J}_j &\leq \mathcal{J}_j(\tilde{u}_j(t|t_{k+1}), \mathbf{e}_j(t_{k+1})) - \mathcal{J}_j(\tilde{u}_j^0(t|t_k), \mathbf{e}_j(t_k)) \\ &= \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1})\|_{\mathbf{Q}_j}^2 + \|\tilde{u}_j(t|t_{k+1})\|_{R_j}^2 dt \\ &\quad + \|\tilde{\mathbf{e}}_j(t_{k+1} + T|t_{k+1})\|_{\mathbf{P}_j}^2 - \|\tilde{\mathbf{e}}_j^0(t_k + T|t_k)\|_{\mathbf{P}_j}^2 \\ &\quad - \int_{t_k}^{t_k+T} \|\tilde{\mathbf{e}}_j^0(t|t_k)\|_{\mathbf{Q}_j}^2 + \|\tilde{u}_j^0(t|t_k)\|_{R_j}^2 dt. \end{aligned} \quad (38)$$

From Lemma 2, it can be seen that

$$\begin{aligned} &\|\tilde{\mathbf{e}}_j(t_{k+1} + T|t_{k+1})\|_{\mathbf{P}_j}^2 - \|\tilde{\mathbf{e}}_j(t_k + T|t_k)\|_{\mathbf{P}_j}^2 \\ &\leq - \int_{t_k+T}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1})\|_{\mathbf{Q}_j}^2 + \|\tilde{u}_j(t|t_{k+1})\|_{R_j}^2 dt. \end{aligned} \quad (39)$$

Substituting (33) and (39) into (38) yields that

$$\Delta \mathcal{J}_j \leq \sum_{i=1}^3 \Delta \mathcal{J}_{ji} \quad (40)$$

where

$$\Delta \mathcal{J}_{j1} = \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1})\|_{\mathbf{Q}_j}^2 - \|\tilde{\mathbf{e}}_j^0(t|t_k)\|_{\mathbf{Q}_j}^2 dt \quad (41a)$$

$$\Delta \mathcal{J}_{j2} = \|\tilde{\mathbf{e}}_j(t_k + T|t_{k+1})\|_{\mathbf{P}_j}^2 - \|\tilde{\mathbf{e}}_j^0(t_k + T|t_k)\|_{\mathbf{P}_j}^2 \quad (41b)$$

$$\Delta \mathcal{J}_{j3} = - \int_{t_k}^{t_{k+1}} \|\tilde{\mathbf{e}}_j^0(t|t_k)\|_{\mathbf{Q}_j}^2 + \|\tilde{u}_j^0(t|t_k)\|_{R_j}^2 dt. \quad (41c)$$

For the first item $\mathcal{J}_{j1}$ in (41a), by using the triangle inequality, it is seen that

$$\begin{aligned} \Delta \mathcal{J}_{j1} &\leq \bar{\lambda}(\mathbf{Q}_j) \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1}) - \tilde{\mathbf{e}}_j^0(t|t_k)\|^2 dt \\ &\quad + 2\bar{\lambda}(\mathbf{Q}_j) \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1}) - \tilde{\mathbf{e}}_j^0(t|t_k)\| \|\tilde{\mathbf{e}}_j^0(t|t_k)\| dt. \end{aligned} \quad (42)$$

By applying the Holder inequality, it is obtained that

$$\begin{aligned} \Delta \mathcal{J}_{j1} &\leq \bar{\lambda}(\mathbf{Q}_j) \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}_j(t|t_{k+1}) - \tilde{\mathbf{e}}_j^0(t|t_k)\| dt \\ &\quad \times \max_{t \in \mathbf{R}_{[t_{k+1}, t_k+T]}} \|\tilde{\mathbf{e}}(t|t_{k+1}) - \tilde{\mathbf{e}}^0(t|t_k)\| \\ &\quad + 2\bar{\lambda}(\mathbf{Q}_j) \int_{t_k+1}^{t_{k+1}+T} \|\tilde{\mathbf{e}}(t|t_{k+1}) - \tilde{\mathbf{e}}^0(t|t_k)\| dt \\ &\quad \times \max_{t \in \mathbf{R}_{[t_{k+1}, t_k+T]}} \|\tilde{\mathbf{e}}^0(t|t_k)\|. \end{aligned} \quad (43)$$

Defining that $\Xi_j \triangleq \max_{t \in \mathbf{R}_{[t_{k+1}, t_k+T]}} \|\tilde{\mathbf{e}}_j(t|t_{k+1}) - \tilde{\mathbf{e}}_j^0(t|t_k)\|$ and $\Psi_j = \max_{t \in \mathbf{R}_{[t_{k+1}, t_k+T]}} \|\tilde{\mathbf{e}}_j^0(t|t_k)\|$, by using $t_{k+1} - t_k \geq \beta_j T$, (42) can be simplified as

$$\Delta \mathcal{J}_{j1} \leq \bar{\lambda}(\mathbf{Q}_j)(1 - \beta_j)T\Xi_j^2 + 2\bar{\lambda}(\mathbf{Q}_j)(1 - \beta_j)T\Xi_j\Psi_j. \quad (44)$$

According to (34), we have that

$$\Xi_j \leq \varpi_j e^{(t_k+T-t_{k+1})} \leq \varpi_j e^{(T-\beta_j T)}. \quad (45)$$

According to (26a), $|\bar{e}_{1j}| < 2\pi$, and $t_{k+1} - t_k \geq \beta_j T$, it can be obtained that

$$\Psi_j \leq \sqrt{(2\pi)^2 + (s_j - \kappa_{1j}|\bar{x}_{d2j}| - \bar{D}_j^* e^{\beta_j T} + \bar{D}_j^*)^2} \triangleq \bar{\Psi}_j. \quad (46)$$

Substituting (45) and (46) into (44) yields that

$$\begin{aligned} \Delta \mathcal{J}_{j1} &\leq \bar{\lambda}(\mathbf{Q}_j)(1 - \beta_j)T\varpi_j e^{(T-\beta_j T)} \\ &\quad (\varpi_j e^{(T-\beta_j T)} + 2\bar{\Psi}_j). \end{aligned} \quad (47)$$

For the second item $\Delta \mathcal{J}_{j2}$ in (41b), we have

$$\begin{aligned} \Delta \mathcal{J}_{j2} &\leq \|\tilde{\mathbf{e}}_j(t_k + T|t_{k+1}) - \tilde{\mathbf{e}}_j^0(t_k + T|t_k)\|_{\mathbf{P}_j} \\ &\quad \times \left(\|\tilde{\mathbf{e}}_j(t_k + T|t_{k+1})\|_{\mathbf{P}_j} + \|\tilde{\mathbf{e}}_j^0(t_k + T|t_k)\|_{\mathbf{P}_j} \right) \\ &\leq \sqrt{\bar{\lambda}(\mathbf{P}_j)} \varpi_j e^{T-\beta_j T} (\gamma_j + \epsilon_j). \end{aligned} \quad (48)$$

For the third item $\Delta \mathcal{J}_{j3}$ in (41c), it is seen that

$$\begin{aligned} \Delta \mathcal{J}_{j3} &\leq - \int_{t_k}^{t_{k+1}} \|\tilde{\mathbf{e}}_j^0(t|t_k)\|_{\mathbf{Q}_j}^2 dt \\ &\leq -\bar{\lambda}(\mathbf{Q}_j)\beta_j T \left(\frac{\gamma_j}{\sqrt{\bar{\lambda}(\mathbf{P}_j)}} - \varpi_j \right)^2. \end{aligned} \quad (49)$$

Therefore, according to (35), by using the same statement as in [20] and [35], it can be proved that the tracking error state will enter the set $\Omega_j(\gamma_j)$ in finite time.

Second, for $\mathbf{e}_j(t_0) \in \Omega_j(\gamma_j)$, as shown in Algorithm 1, the control law is switched to $\mathbf{K}_j \mathbf{e}_j(t)$. Taking the terminal cost function $\mathcal{V}_j(\cdot)$ as the Lyapunov function, it can be seen that $\mathcal{V}_j(\mathbf{e}_j(t)) \leq -\|\mathbf{e}_j(t)\|_{\mathbf{Q}_j^*}^2 + 2\mathbf{e}_j(t)^T \mathbf{P}_j \tilde{\mathbf{d}}^*(t) \leq (\bar{\lambda}(\mathbf{Q}_j^*)/\bar{\lambda}(\mathbf{P}_j))(-\|\mathbf{e}_j(t)\|_{\mathbf{P}_j}^2 + (2\bar{\lambda}(\mathbf{P}_j)\|\sqrt{\mathbf{P}_j}\|/\bar{\lambda}(\mathbf{Q}_j^*))\gamma_j \bar{D}_j^*)$. Then, by using the same argument in [20] and [21], it follows (36) that, $\mathbf{e}_j(t)$ will remain stable in the robust invariant set $\Omega_j(\gamma_j)$.

Therefore, the proof is completed by discussing the above two cases. ■

Remark 7: This theorem implies that the closed-loop stability can be ensured by properly choosing L_j , β_j , and T . Referring to Remark 6, the tradeoff between control performance and computational resources should be carefully considered in application practice when designing these parameters.

V. SIMULATION STUDY

In this section, simulation experiments for a PUMA 560 manipulator are conducted in MATLAB on a laptop with a processor running at 2.21 GHz and 8.0-GB RAM. The sampling interval and simulation time are set to 0.05 and 20 s, respectively. The piecewise-constant control law is employed over the sampling intervals in this simulation.

A. Modeling Parameters of PUMA 560

The modeling parameters of the manipulator are referring to the actual measured geometric and inertial parameters of the PUMA 560 in [37] and [38]. The physical limits of the manipulator are given in Table II [37]. As reported in [38],

TABLE II
PHYSICAL LIMITS OF PUMA 560

Joint j	1	2	3	4	5	6
s_{2j} (rad/s)	8	10	10	5	5	5
c_j (N·m)	97.6	186.4	89.4	24.2	20.1	21.3

TABLE III
ESTIMATES OF ξ_{ij}, h_j

Parameter	Value	Parameter	Value	Parameter	Value
ξ_{11}	0.24	ξ_{12}	0.41	ξ_{13}	0.46
ξ_{14}	0.20	ξ_{15}	0.19	ξ_{16}	0.12
ξ_{21}	7.91	ξ_{22}	2.89	ξ_{23}	8.17
ξ_{24}	6.66	ξ_{25}	5.99	ξ_{26}	3.43
h_1	27.55	h_2	56.11	h_3	13.04
h_4	3.29	h_5	2.07	h_6	1.81

there would exist measurement errors in the mechanical and dynamic parameters of the manipulator. Referring to [2], in this simulation, disturbances are introduced by randomly perturbing the measured masses of all links and the friction coefficients of all joints by $\pm 3\%$. ζ_j is set to the j th diagonal element of $M(0)$. By applying the estimation approach in [39] to the 100 random trajectories each of which runs 20 s, ξ_{ij} and h_j are estimated (Table III). Note that more accurate estimation of the above parameters would be required to further improve the robustness of the controller in application practice, which is not discussed herein.

B. Comparative Simulations

The DO-ET-RMPC algorithm is implemented in comparison with the DO-RMPC, RMPC, ET-RMPC, and filtered integral sliding-mode-based RMPC (FISM-RMPC) algorithms to verify the effectiveness. All algorithms are implemented on the tracking error subsystems (8) decoupled by the CTLC approach. Two simulation cases are provided as follows.

1) *Case 1*: The control objective in this case is to track the quintic polynomial trajectories in the joint space from the static initial angular displacements $[-\pi/3, 0, \pi/4, 0, -\pi/3, -\pi/4]^T$ to the static terminal ones $[0, \pi/4, 0, \pi/3, 0, 0]^T$. The initial angular displacements and velocities of the PUMA 560 are chosen as $[\pi/3, -\pi/2, 3\pi/4, -\pi/3, \pi/6, \pi/4]^T$ and $[-3.74, -4.61, 4.57, -2.28, 2.29, 2.36]^T$, respectively. The overshoot factors are set as $\kappa_{1j} = \kappa_{2j} = 1$. The parameters of the above five algorithms are set as follows.

a) *DO-ET-RMPC algorithm*: According to Lemma 2 and Theorems 2 and 3, the MPC parameters of the DO-ET-RMPC algorithm are as follows: $T = 1$ s, $L_j = -100$, $\beta_1 = \beta_2 = \beta_3 = 0.7$, $\beta_4 = \beta_5 = \beta_6 = 0.9$, $j \in \mathbf{I}_{[1,6]}$, $\mathbf{Q}_1 = \text{diag}\{0.1, 0.1\}$, $\mathbf{Q}_2 = \mathbf{Q}_3 = \text{diag}\{0.2, 0.2\}$, $\mathbf{Q}_4 = \mathbf{Q}_5 = \mathbf{Q}_6 = \text{diag}\{0.3, 0.3\}$, $R_1 = R_2 = 0.02$, $R_3 = 0.01$, $R_4 = R_5 = R_6 = 0.02$, $\mathbf{K}_1 = [2.2361 \ 3.0777]$, $\mathbf{K}_2 = [3.1623 \ 4.0404]$, $\mathbf{K}_3 = [4.4721 \ 5.3800]$, $\mathbf{K}_4 = \mathbf{K}_5 = \mathbf{K}_6 = [3.8730 \ 4.7693]$, $\mathbf{P}_1 = \begin{bmatrix} 0.1376 & 0.0447 \\ 0.0447 & 0.0616 \end{bmatrix}$, $\mathbf{P}_2 = \begin{bmatrix} 0.2555 & 0.0632 \\ 0.0632 & 0.0808 \end{bmatrix}$, $\mathbf{P}_3 = \begin{bmatrix} 0.2406 & 0.0447 \\ 0.0447 & 0.0538 \end{bmatrix}$, $\mathbf{P}_4 = \mathbf{P}_5 = \mathbf{P}_6 = \begin{bmatrix} 0.3694 & 0.0775 \\ 0.0775 & 0.0954 \end{bmatrix}$, $\gamma_1 = 1.2$, $\gamma_2 = \gamma_3 = 1.9$, $\gamma_4 = \gamma_5 = \gamma_6 = 0.9$, $\epsilon_1 = 1.0681$, $\epsilon_2 = 1.6056$, $\epsilon_3 = 1.5849$, $\epsilon_4 = \epsilon_5 = \epsilon_6 = 0.6973$.

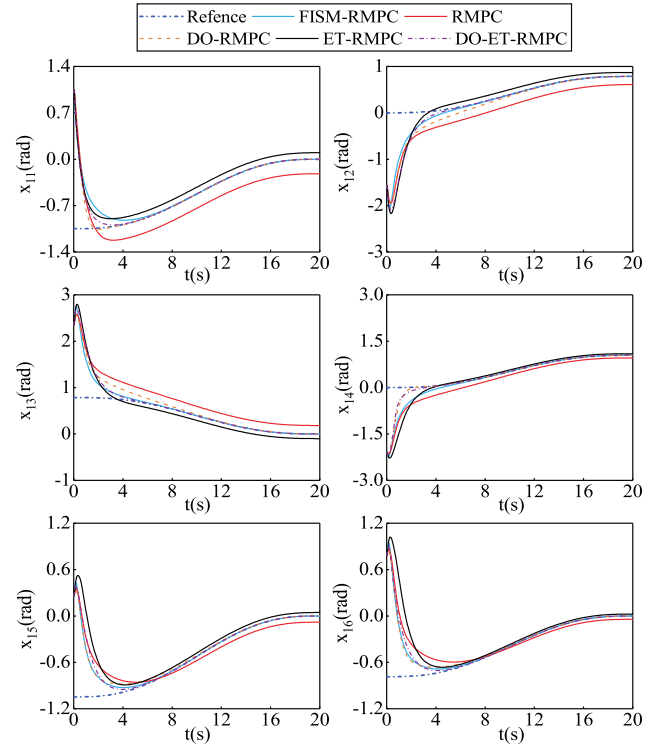


Fig. 3. Comparison of actual tracking trajectories ($x_{1j}(t), j \in \mathbf{I}_{[1,6]}$).

b) *ET-RMPC algorithm*: For simplicity, the ET-RMPC algorithm is implemented by directly setting $u_j^{do} = -\hat{d}_j^* = 0$ and $\tilde{C}_{ej} = C_{ej}$. It can be easily verified to be adapted to the MPC parameter environment of the DO-ET-RMPC algorithm.

c) *DO-RMPC algorithm* [36]: The parameters of the DO-RMPC algorithm have been calibrated to our best effort. The best set of parameters during the calibration might be as follows: $T = 1$ s, $L_j = -100$, $j \in \mathbf{I}_{[1,6]}$, $\mathbf{Q}_1 = \text{diag}\{1.5, 1.5\}$, $\mathbf{Q}_2 = \mathbf{Q}_3 = \text{diag}\{3, 3\}$, $\mathbf{Q}_4 = \mathbf{Q}_5 = \mathbf{Q}_6 = \text{diag}\{4, 4\}$, $R_1 = 0.2$, $R_2 = R_3 = 0.3$, $R_4 = R_5 = R_6 = 0.4$, $\mathbf{K}_1 = [2.7386 \ 3.6024]$, $\mathbf{K}_2 = \mathbf{K}_3 = [3.1623 \ 4.0404]$, $\mathbf{K}_4 = \mathbf{K}_5 = \mathbf{K}_6 = [3.1623 \ 4.0404]$, $\mathbf{P}_1 = \begin{bmatrix} 1.9731 & 0.5477 \\ 0.5477 & 0.7205 \end{bmatrix}$, $\mathbf{P}_2 = \begin{bmatrix} 3.8330 & 0.9487 \\ 0.9487 & 1.2121 \end{bmatrix}$, $\mathbf{P}_3 = \mathbf{P}_4 = \mathbf{P}_5 = \mathbf{P}_6 = \begin{bmatrix} 5.1107 & 1.2649 \\ 1.2649 & 1.6161 \end{bmatrix}$, $\gamma_1 = 3$, $\gamma_2 = 7$, $\gamma_3 = 8$, $\gamma_4 = \gamma_5 = \gamma_6 = 4$, $\epsilon_1 = 2.9562$, $\epsilon_2 = 6.8895$, $\epsilon_3 = 7.8760$, $\epsilon_4 = \epsilon_5 = \epsilon_6 = 3.9373$.

d) *RMPC algorithm*: As in [36], the RMPC algorithm without disturbance compensation is implemented based on the same MPC parameters as in the DO-RMPC algorithm, with $\tilde{C}_{ej} = C_{ej}$.

e) *FISM-RMPC algorithm* [33]: The time constant of the filter and the gain of the unit vector approach in [33] are set as 0.3 and $1.1\xi_{1j}$, respectively. The MPC parameters are also chosen as same as the DO-RMPC algorithm with that $\tilde{S}_{ej} = S_{ej}$ and $\tilde{C}_{ej} = \{\tilde{u}_j \in \mathbf{R} | |\tilde{u}_j| \leq v_j - \kappa_{2j}|u_{dj}| - 1.1\xi_{1j}, j \in \mathbf{I}_{[1,n]}\}$.

f) *Results and analysis*: Fig. 3 shows the actual tracking trajectories under the above five algorithms. The tracking error trajectories are illustrated in Fig. 4. Fig. 5 further details the evolution of the tracking errors under the FISM-RMPC, DO-RMPC, and DO-ET-RMPC algorithms after 10 s. The control torques are exhibited in Fig. 6, while the disturbance

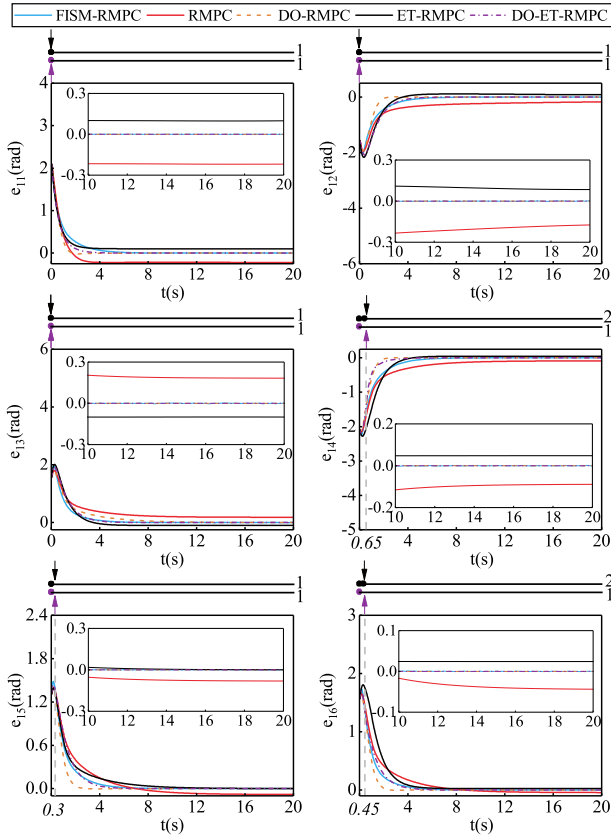


Fig. 4. Comparison of tracking error trajectories ($e_{1j}(t), j \in \mathbf{I}_{[1,6]}$). Solid circle: triggering time instant. Arrows: switching time instant. Purple: DO-ET-RMPC. Black: ET-RMPC.

estimation errors of the DO-RMPC and DO-ET-RMPC algorithms are shown in Fig. 7. In addition, the root mean-square tracking error (RMSE) and root mean-square control input (RMSC), as mentioned in [33], are introduced to quantitatively evaluate average tracking performance. In particular, RMSE^{10} denotes RMSE calculated on [10, 20] s. T_t represents the triggering times while P_{pd} stands for the percentage of the RMSE value increase of the DO-ET-RMPC algorithm over the DO-RMPC algorithm. The comparison results of the quantitative performance indexes are presented in Table IV and graphically rendered in Fig. 8.

In particular, from Figs. 4 and 5, under the selected parameter environments, it can be seen that the steady-state errors under the FISM-RMPC, DO-RMPC, and DO-ET-RMPC algorithms are obviously smaller than those under the RMPC and ET-RMPC algorithms, which could be credited to the continuous active disturbance compensation generated by the disturbance observer or FISM controller. From Fig. 8 and Table IV, it is seen that the RMSE (RMSE^{10}) values under the FISM-RMPC, DO-RMPC, and DO-ET-RMPC algorithms are smaller than those under the RMPC and ET-RMPC algorithms for all joints. For instance, for joint 1, the RMSE values 0.3196 (FISM-RMPC), 0.2910 (DO-RMPC), and 0.2871 (DO-ET-RMPC) are smaller than the RMSE values 0.3648 (RMPC) and 0.3375 (ET-RMPC), and the RMSE^{10} values 0.0335 (FISM-RMPC), 4.3×10^{-5} (DO-RMPC), and 2.2×10^{-5} (DO-ET-RMPC) are smaller than the RMSE^{10}

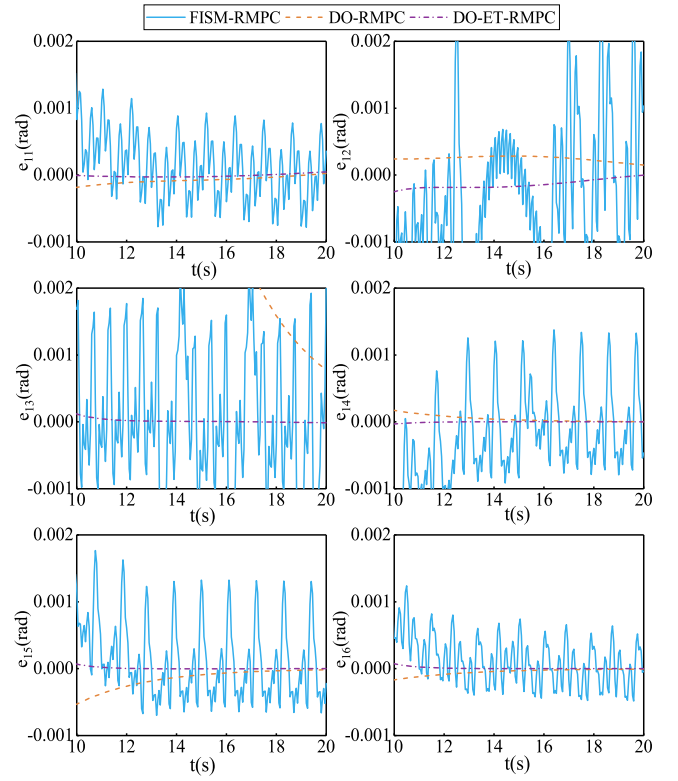


Fig. 5. Comparison of tracking error trajectories ($e_{1j}(t), j \in \mathbf{I}_{[1,6]}$) under FISM-RMPC, DO-RMPC, and DO-ET-RMPC from 10 to 20 s.

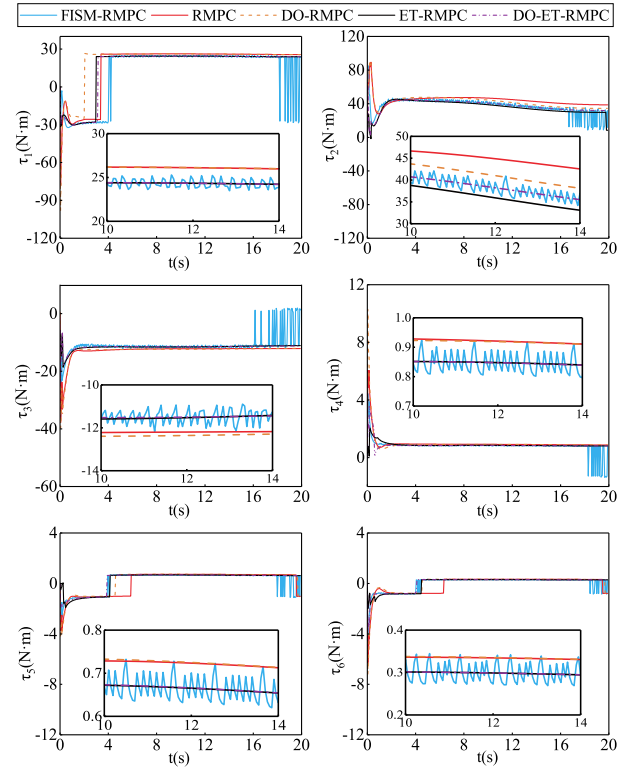
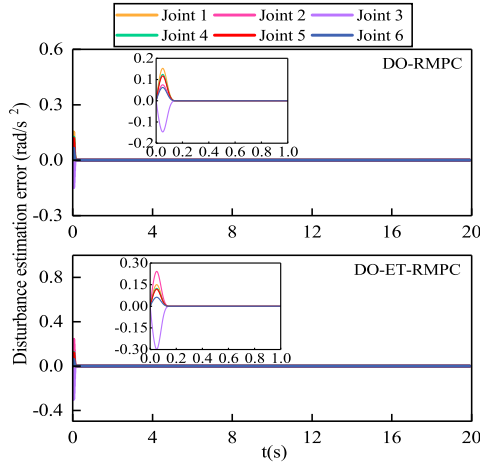


Fig. 6. Comparison of control torques ($\tau_j(t), j \in \mathbf{I}_{[1,6]}$).

values 0.2149 (RMPC) and 0.0975 (ET-RMPC). These results would demonstrate the benefits of the active disturbance rejection techniques through the quantitative evaluation of

Fig. 7. Disturbance estimation errors ($\tilde{d}_j^*(t), j \in \mathbf{I}[1,6]$).TABLE IV
COMPARISON OF QUANTITATIVE PERFORMANCE OF FIVE ALGORITHMS

Joints	Algorithms	RMSE (rad)	RMSC (N·m)	P_{pd}	$\text{RMSE}^{10}(\text{rad})$	T_t
1	FISM-RMPC	0.3196	25.0833	-	0.0335	400
	RMPC	0.3648	26.3556	-	0.2149	400
	DO-RMPC	0.2910	26.3644	-	4.3×10^{-5}	400
	ET-RMPC	0.3375	24.7941	-	0.0975	1
	DO-ET-RMPC	0.2871	24.8552	-1.34%	2.2×10^{-5}	1
2	FISM-RMPC	0.4032	38.6978	-	0.0226	400
	RMPC	0.4973	44.4786	-	0.1988	400
	DO-RMPC	0.4439	43.0495	-	1.3×10^{-4}	400
	ET-RMPC	0.4961	36.8134	-	0.0939	1
	DO-ET-RMPC	0.4621	38.2697	4.10%	1.5×10^{-4}	1
3	FISM-RMPC	0.3618	11.5743	-	0.0182	400
	RMPC	0.4631	13.222	-	0.1855	400
	DO-RMPC	0.3904	13.3191	-	3.3×10^{-5}	400
	ET-RMPC	0.4325	11.6888	-	0.0981	1
	DO-ET-RMPC	0.4066	11.6577	4.15%	2.8×10^{-5}	1
4	FISM-RMPC	0.4276	0.9554	-	0.0071	400
	RMPC	0.4732	1.1451	-	0.0948	400
	DO-RMPC	0.3662	1.2641	-	4.2×10^{-5}	400
	ET-RMPC	0.5188	0.8943	-	0.0471	2
	DO-ET-RMPC	0.3679	0.9413	0.46%	2.2×10^{-5}	1
5	FISM-RMPC	0.2845	0.8380	-	0.0072	400
	RMPC	0.3149	0.9410	-	0.0728	400
	DO-RMPC	0.2773	0.9420	-	4.1×10^{-5}	400
	ET-RMPC	0.3772	0.8027	-	0.0440	1
	DO-ET-RMPC	0.2895	0.8099	4.40%	1.9×10^{-5}	1
6	FISM-RMPC	0.3188	0.6257	-	0.0066	400
	RMPC	0.3546	0.7776	-	0.0365	400
	DO-RMPC	0.3106	0.7971	-	4.6×10^{-5}	400
	ET-RMPC	0.4297	0.5263	-	0.0239	2
	DO-ET-RMPC	0.3246	0.5773	4.51%	2.1×10^{-5}	1

average tracking performance. However, as observed from Figs. 5 and 6, compared with the DO-RMPC and DO-ET-RMPC algorithms, there exists inevitable chattering when using the FISM-RMPC algorithm. Although the first-order linear filter in the FISM controller can alleviate oscillations, *ad hoc* parameter tuning in the filter would be required to avoid sacrificing much anti-disturbance performance of the FISM controller. On the contrary, the disturbance observer, with a simpler parameter design, could provide smooth control input actions while also guaranteeing satisfactory disturbance compensation.

Further, it can be seen from Figs. 3 and 4 that the initial response speed (for $t < 2$ s) of the DO-ET-RMPC algorithm would perform slower than the DO-RMPC algorithm for

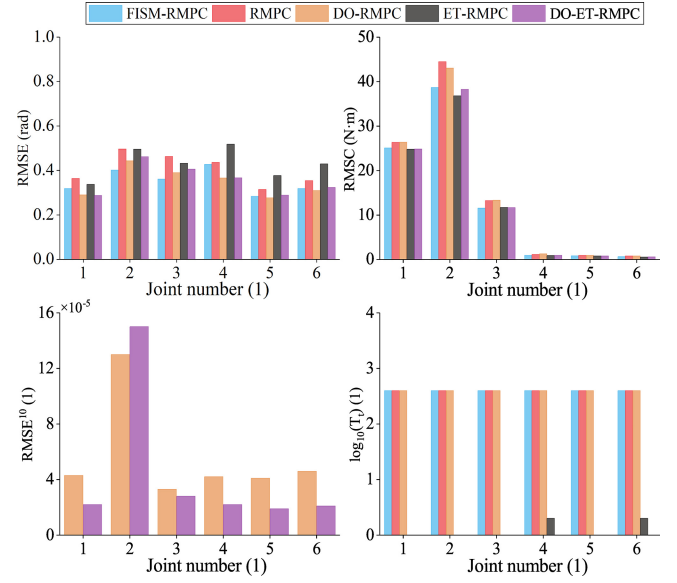


Fig. 8. Comparison of quantitative performance of five algorithms.

almost all joints, which would result in a larger RMSE value for the DO-ET-RMPC algorithm. It can be seen from Table IV that the percentages of the average tracking performance degradation of the DO-ET-RMPC algorithm over the DO-RMPC algorithm for the six joints are -1.34%, 4.10%, 4.15%, 0.46%, 4.40%, and 4.51%, respectively. They are all smaller than 5% in this simulation case, which would imply that the DO-ET-RMPC and DO-RMPC algorithms are comparable. In addition, as shown in Fig. 5, the tracking error trajectories of the DO-RMPC and DO-ET-RMPC algorithms would be very close after 10 s. Denote the deviation of the RMSE^{10} values of the DO-RMPC and DO-ET-RMPC algorithms as RMSE_d^{10} . From Table IV and Fig. 8, it is easily calculated that the maximal RMSE_d^{10} value among the six joints is lesser than 2.5×10^{-5} rad, which would imply that the two algorithms would also have comparable steady-state performance. Further, it is worth mentioning that the T_t value of the DO-ET-RMPC algorithm is greatly lesser than that of the DO-RMPC algorithm, for example, $1 < 400$ for the joint 1. Therefore, by summarizing the above results, it could be concluded that the DO-ET-RMPC algorithm could achieve a better trade-off between computational resources and control performance. Moreover, it can be also witnessed from Table IV that, in addition to the advantage in tracking performance, the DO-ET-RMPC algorithm also uses fewer control updates than the ET-RMPC algorithm for joints 4 and 6, which would imply, as analyzed before, that the continuous disturbance compensation could potentially reduce the triggering times while improving the tracking performance effectively.

2) Case 2: To validate the reliability of the above results, an additional ten reference trajectories are chosen randomly from the above 100 trajectories for further comparative experiments of the DO-RMPC, ET-RMPC, and DO-ET-RMPC algorithms. The parameter settings of these three algorithms are as same as in case 1. The statistical results of the quantitative performance values (P_{pd} and RMSE^{10}) are presented in Table V. Fig. 9 counts the T_t values of the ET-RMPC

TABLE V
COMPARISON OF QUANTITATIVE PERFORMANCE OF DO-RMPC,
ET-RMPC, AND DO-ET-RMPC ALGORITHMS

Joints	Algorithms		P_{pd}	RMSE ¹⁰ (rad)	RMSE _d ¹⁰ (rad)
1	DO-RMPC	Max.	-	6.2×10^{-5}	-
		Min.	-	2.2×10^{-5}	-
		mean	-	4.1×10^{-5}	-
	ET-RMPC	Max.	-	0.1099	-
		Min.	-	0.0981	-
		mean	-	0.1041	-
	DO-ET-RMPC	Max.	2.36%	3.2×10^{-5}	3.2×10^{-5}
		Min.	-1.68%	0.4×10^{-5}	1.2×10^{-5}
		mean	-0.21%	1.3×10^{-5}	2.2×10^{-5}
2	DO-RMPC	Max.	-	4.4×10^{-4}	-
		Min.	-	0.5×10^{-4}	-
		mean	-	2.9×10^{-4}	-
	ET-RMPC	Max.	-	0.1176	-
		Min.	-	0.0562	-
		mean	-	0.0868	-
	DO-ET-RMPC	Max.	6.89%	3.1×10^{-4}	0.2×10^{-5}
		Min.	3.99%	0.5×10^{-4}	-0.8×10^{-5}
		mean	5.59%	1.8×10^{-4}	-0.3×10^{-5}
3	DO-RMPC	Max.	-	8.2×10^{-4}	-
		Min.	-	2.2×10^{-5}	-
		mean	-	7.3×10^{-4}	-
	ET-RMPC	Max.	-	0.0391	-
		Min.	-	0.0083	-
		mean	-	0.0259	-
	DO-ET-RMPC	Max.	6.72%	4.4×10^{-4}	3.8×10^{-5}
		Min.	-3.25%	1.5×10^{-5}	0.6×10^{-5}
		mean	2.89%	2.9×10^{-4}	2.8×10^{-5}
4	DO-RMPC	Max.	-	9.2×10^{-5}	-
		Min.	-	3.8×10^{-5}	-
		mean	-	7.2×10^{-5}	-
	ET-RMPC	Max.	-	0.0715	-
		Min.	-	0.0471	-
		mean	-	0.0593	-
	DO-ET-RMPC	Max.	1.31%	3.4×10^{-5}	3.3×10^{-5}
		Min.	0.19%	5.6×10^{-6}	0.4×10^{-5}
		mean	0.69%	1.7×10^{-5}	1.4×10^{-5}
5	DO-RMPC	Max.	-	6.6×10^{-5}	-
		Min.	-	2.7×10^{-5}	-
		mean	-	4.2×10^{-5}	-
	ET-RMPC	Max.	-	0.0705	-
		Min.	-	0.0427	-
		mean	-	0.0637	-
	DO-ET-RMPC	Max.	6.82%	2.3×10^{-5}	4.4×10^{-5}
		Min.	2.63%	9.4×10^{-6}	1.2×10^{-5}
		mean	5.84%	1.5×10^{-5}	2.4×10^{-5}
6	DO-RMPC	Max.	-	5.5×10^{-5}	-
		Min.	-	1.8×10^{-5}	-
		mean	-	3.4×10^{-5}	-
	ET-RMPC	Max.	-	0.0632	-
		Min.	-	0.0238	-
		mean	-	0.0593	-
	DO-ET-RMPC	Max.	6.93%	5.3×10^{-5}	2.1×10^{-5}
		Min.	4.43%	0.3×10^{-5}	1.4×10^{-5}
		mean	5.01%	3.3×10^{-5}	1.6×10^{-5}

and DO-ET-RMPC algorithms. In this simulation case, the maximal P_{pd} values for the six joints are 2.36%, 6.89%, 6.72%, 1.31%, 6.82%, and 6.93%, respectively. They are all smaller than 7%. In addition, the RMSE¹⁰ values under the DO-ET-RMPC algorithm for all six joints are all still close, respectively, to those under the DO-RMPC algorithm (i.e., the maximal RMSE_d¹⁰ values for all joints are all smaller than 5×10^{-5} rad) and are greatly smaller than those under the ET-RMPC algorithm, which also implies that both the DO-ET-RMPC and DO-RMPC algorithms are comparable in steady-state performance. More importantly, for each joint, the

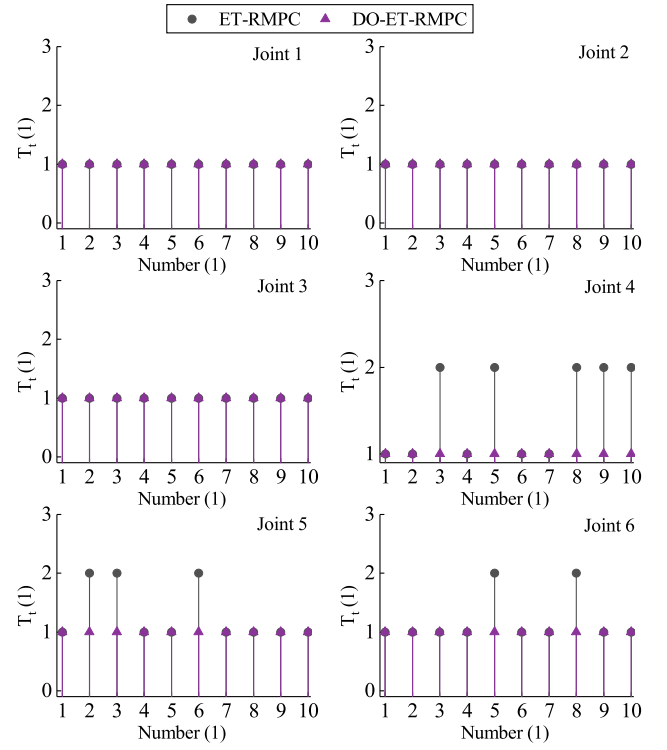


Fig. 9. Comparison of T_t under ET-RMPC and DO-ET-RMPC algorithms in ten simulation experiments.

T_t value under the DO-ET-RMPC algorithm would be significantly lesser than that under the DO-RMPC algorithm and slightly lesser than that by the ET-RMPC algorithm. Therefore, in comparison with the regular DO-RMPC and ET-RMPC algorithms, the DO-ET-RMPC algorithm could achieve a better tradeoff between control performance and computational resources, thanks to the organic integration of the disturbance observer technique and event-triggering mechanism into MPC.

3) *Practical Consideration in Parameter Design:* From Remarks 6 and 7, the satisfaction of Theorems 2 and 3 highly relies on the parameters β_j , T , and $\bar{D}_j^*$. The design of the parameter β_j needs a tradeoff between control performance and computational resources. In addition, as stated in [20], [21], and [26], the upper bound of the maximum-allowable disturbances of the ET-MPC algorithm could be estimated conservatively under a set of deterministic parameters β_j , T . Generally, a larger β_j , as well as a smaller T , would allow the proposed control scheme to endure a larger disturbance (i.e., a larger $\bar{D}_j^*$) [21]. Note that, to guarantee strict constraint satisfaction, a too-large disturbance might cause the feasibility and stability of the proposed control scheme would not to be guaranteed within a limited range of parameter adjustments [e.g., $\beta_j \in \mathbf{R}_{(0,1)}$].

VI. CONCLUSION

In this study, a novel hierarchical control framework combining CTLC with DO-ET-RMPC is proposed for the trajectory tracking control of uncertain robotic manipulators subject to state and input constraints. Specifically, by using the CTLC approach, a set of decoupling linear tracking error subsystems are derived to reduce the optimization complexity of

MPC. Then the DO-ET-RMPC algorithm is developed to stabilize these subsystems. The corresponding continuous-time theoretical properties of the proposed control scheme, considering disturbances and state and control input constraints, are provided for the first time, including the avoidance of Zeno behavior, robust constraint satisfaction, recursive feasibility, and closed-loop stability. The effectiveness of the proposed control scheme is verified by the comparative simulations. In particular, from the statistical quantitative results, the DO-ET-RMPC algorithm uses at most one control update compared with the DO-RMPC algorithm which uses 400 control updates, and the ET-RMPC algorithm which uses at most two control updates. Meanwhile, for all joints, the maximal P_{pd} values and the maximal $RMSE_d^{10}$ values are smaller than 7% and 5×10^{-5} rad, respectively. The DO-ET-RMPC algorithm achieves a satisfactory tradeoff between control performance and computational resources. It is worth also mentioning that the proposed control algorithm would be applicable to other robotic systems which have the similar dynamics, such as unmanned surface vehicles.

Furthermore, the conservativeness of the tightened constraint sets in the DO-ET-RMPC algorithm would be caused by the constant estimation of the residual disturbance upper bound. It would be imperative to reduce the conservativeness to improve the robustness against disturbances, and time-varying MPC might be a clue. Future studies would focus on the reduction of the conservativeness, and the extension of the above results to the general nonlinear constrained uncertain systems.

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